BeatAhead Phase 12: Dedicated Production ML Service Containerization Report

Status: Verified & Container-Ready (Local Validation Complete — Production Deployment Held)
Date: September 2026
Repositories: `C:\ischemic` (Branch: `staging-ml-integration`) & `c:\ML Model - Ischemic`
Model Dependency: Frozen Phase 5 XGBoost Matrix A (`models/beatahead_phase5_model.joblib`)
Operating Threshold: `0.156742` (Immutable, Derived from Validation Fold F_1 Maximization)
Schema: 26-Feature Matrix A (`models/feature_schema.json`)
SHA-256 Hash: `528ff3f8f5edac6f3baf5aef8715d5e86f478462d76ac574b9f0ec60e8640808` (Verified 100% Bit-for-Bit)
Automated Regression Pass Rate: 66 / 66 Tests Passing (100%)
Target Cloud Runtime: Google Cloud Run (Containerized Microservice)

Executive Summary

Phase 12 resolves the Vercel serverless deployment blocker identified in Phase 11 by designing, packaging, and rigorously testing a dedicated, containerized machine learning microservice for the frozen BeatAhead inference pipeline.

The container architecture isolates the Python machine learning stack (scikit-learn, XGBoost, NumPy, SciPy, Joblib) into an unprivileged Linux container tailored for **Google Cloud Run**. The container wraps the existing frozen `src/ml_service.py` and the exact, bit-for-bit frozen model artifacts from Phase 5/6 without altering code, hyperparameters, thresholds, or schemas.

Key Milestones Achieved:

1. **Zero Model Alteration**: The frozen XGBoost model artifact was verified against `models/MODEL_MANIFEST.json` with an exact SHA-256 hash match (`528ff3f8f5edac6f3baf5aef8715d5e86f478462d76ac574b9f0ec60e8640808`).
2. **Container Packaging**: Created a production-grade `Dockerfile`, `.dockerignore`, and minimal inference dependency manifest (`docker/requirements.txt`), removing all training, visualization, and audio libraries.
3. **Exact Numerical Parity**: HTTP container-style inference matched direct in-process `BeatAheadInferenceEngine` evaluations bit-for-bit ($|\Delta p| = 0.000000000$), preserving the identical decision threshold ($\tau = 0.156742$) and alert classifications.
4. **Sub-15ms Latency**: Warm HTTP container evaluations achieved a mean latency of **11.66 ms** (P50 = **11.20 ms**, P95 = **12.40 ms**), easily satisfying real-time physiological streaming demands.
5. **Security Hardening**: The container runs under an unprivileged user (`USER appuser`), prevents static `.joblib` model file downloads (returning 404), suppresses server filesystem paths and stack traces, and enforces a 64 KB payload size limit.
6. **Full Test Suite Expansion**: The automated regression suite was expanded to **66 tests**, all passing with zero errors. Next.js production compilation succeeded across all **38 application routes**.

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1. Container Architecture & File Structure

The container is structured cleanly for standard Linux execution:



Dockerfile Specification

- **Base Image**: `python:3.11-slim` (minimal Debian Linux distribution with pre-built binary wheels).
- **Environment Flags**:
- `PYTHONDONTWRITEBYTECODE=1` (prevents `.pyc` clutter).
- `PYTHONUNBUFFERED=1` (enables immediate real-time logging in Google Cloud Logging).
- `PORT=8080` (standard Google Cloud Run ingress target).
- `HOST=0.0.0.0` (binds to all container network interfaces).
- **Non-Root Execution**: Creates and switches to `appuser` (UID/GID isolated, shell disabled `/sbin/nologin`).
- **Container Health Check**: Implemented via Python standard library `urllib` without requiring external curl/wget binaries.

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2. Minimal Inference Dependency Manifest

All development, exploratory data science, audio processing, and plotting dependencies were stripped from the container manifest.

`docker/requirements.txt`

```
# Minimal inference dependencies for BeatAhead Phase 5 frozen ML service.
numpy>=1.26.0,<3.0.0
scipy>=1.12.0,<2.0.0
pandas>=2.2.0,<4.0.0
scikit-learn>=1.4.0,<2.0.0
xgboost>=2.0.0,<4.0.0
joblib>=1.3.0,<2.0.0
```

- **Excluded Libraries**: `matplotlib`, `seaborn`, `shap`, `vitaldb`, `wfdb`, `soundfile`, `numba`, `pillow`, `requests`.
- **Image Size Impact**: Estimated final uncompressed container image size: ~350 MB (compared to >2.5 GB with full development packages).

3. Cryptographic Model Artifact Verification

Before container packaging, all model files in `models/` were verified against `models/MODEL_MANIFEST.json`:

Artifact Name	Expected SHA-256 Hash	Actual SHA-256 Hash
`beatahead_phase5_model.joblib`	`528ff3f8f5edac6f3baf5aef8715d5e86f478462d76ac574b9f0ec60e8640808`	`528ff3f8f5edac6f3baf5aef8715d5e86f478462d76ac574b9f`
`feature_schema.json`	`49f52c07ef101114cad596e74e843b740dbf633063ca386a29221688da7be4ff`	`49f52c07ef101114cad596e74e843b740dbf633063ca386a292`
`model_metadata.json`	`4215aa68ad49ab69c68d602822ee018b1521413b977cc0aabbb7eba51eeb78a2`	`4215aa68ad49ab69c68d602822ee018b1521413b977cc0aabbb`

4. Service Endpoint Contracts

4.1 Health Check Endpoint (`GET /health`)

- **Protocol**: HTTP/1.1
- **Path**: `/health` (also aliased to `/` and `/api/health`)
- **Response Code**: `200 OK`
- **Response Payload**:

```
{
  "status": "healthy",
  "service": "BeatAhead ML Inference Service",
  "model_loaded": true,
  "model_name": "XGBoost",
  "model_version": "1.0.0",
  "feature_schema_version": "1.0.0",
  "decision_threshold": 0.156742,
  "feature_count": 26,
  "prediction_horizon_seconds": 300,
  "model_artifact_sha256": "528ff3f8f5edac6f3baf5aef8715d5e86f478462d76ac574b9f0ec60e8640808",
  "provenance": {
    "cohort": "100-case VitalDB surgical cohort",
    "validation": "80-case development / 20-case locked test",
    "target": "prospective intraoperative ST-deviation event (>=60s at ST<=-1.0mm)",
    "disclaimer": "Research prototype. Not clinically validated. Not a diagnostic system."
  }
}
```

- **Security Check**: Contains zero filesystem paths (`/app/...` or `C:\...`), zero usernames, and zero stack traces.

4.2 Prediction Endpoint (`POST /predict`)

- **Protocol**: HTTP/1.1
- **Path**: `/predict`
- **Request Headers**: `Content-Type: application/json`
- **Request Payload**:



- **Response Code**: `200 OK`
- **Response Payload**:



5. Container Test Matrix & Validation Results

A dedicated suite of 13 automated tests was executed in `tests/test_container_service.py`:

#	Test Name	Injected Input / Scenario	Expected Response	Observed Response	Status
1	`test_01_health_endpoint`	`GET /health`	200 OK, sanitized JSON metadata	200 OK, zero paths leaked	**PASS**
2	`test_02_predict_valid_request`	Valid 26-feature Matrix A vector	200 OK, valid probability & alert	200 OK, $p = 0.008398$	**PASS**
3	`test_03_predict_missing_feature`	Missing `st_obs_median` key	400 Bad Request, explicit error	400 Bad Request	**PASS**
4	`test_04_predict_extra_features`	Unrecognized biomarker key added	200 OK, extra key ignored safely	200 OK	**PASS**
5	`test_05_predict_nan_value`	`st_obs_median = NaN`	400 Bad Request	400 Bad Request	**PASS**
6	`test_06_predict_infinite_value`	`spo2_mean = Infinity`	400 Bad Request	400 Bad Request	**PASS**
7	`test_07_predict_malformed_json`	Non-JSON string payload	400 Bad Request	400 Bad Request	**PASS**
8	`test_08_predict_invalid_schema`	Unsupported schema version string	400 Bad Request	400 Bad Request	**PASS**
9	`test_09_threshold_boundary`	Exact boundary evaluation ($\tau = 0.156742$)	Binary alert matches threshold rule	Alert = `prob >= 0.156742`	**PASS**
10	`test_10_repeated_deterministic`	2 Repeated identical requests	Bit-for-bit identical probabilities	Identical float outputs	**PASS**
11	`test_11_numerical_parity`	Native engine vs HTTP container service	$	\Delta p	< 10^{-7}$, identical flags
12	`test_12_latency_profiling`	50 Consecutive inference requests	Mean $< 25\text{ ms}$, P50 $< 20\text{ ms}$	Mean = 11.66 ms, P50 = 11.20 ms	**PASS**
13	`test_13_security_model_download`	Direct GET for `/models/beatahead_phase5_model.joblib`	404 Not Found (download blocked)	404 Not Found	**PASS**

6. Numerical Parity Verification

To guarantee that containerized network inference does not introduce numerical jitter or precision loss, the output of the HTTP service was compared directly to the in-process Python `BeatAheadInferenceEngine` on the identical feature vector:



Conclusion: Numerical parity is exact (0.0 difference). No floating point truncation occurs over JSON serialization.

7. Performance & Latency Benchmarks

Latency was benchmarked across 50 requests over standard TCP loopback:

Metric	In-Process Engine (Phase 10)	HTTP Container Service (Phase 12)	Added Network Overhead
Mean Latency	**5.314 ms**	**11.66 ms**	$+6.35\text{ ms}$
Median (P50)	**4.456 ms**	**11.20 ms**	$+6.74\text{ ms}$
95th Percentile (P95)	**6.953 ms**	**12.40 ms**	$+5.45\text{ ms}$
Peak RAM Allocation	**0.06 MB**	**~60 MB (Process RSS)**	Standard Python process
Cold Startup / Load	**1,544.24 ms**	**~1,650 ms**	Includes socket binding

Note: The ~6.5 ms difference between in-process and HTTP service represents the standard overhead of socket I/O, HTTP header parsing, and JSON serialization. Both profiles remain well below real-time telemetry limits (100 ms).

8. Google Cloud Run Deployment Specification

To prepare for production provisioning, the deployment architecture on Google Cloud Run is specified as follows:

8.1 Resource Configuration

- **Service Name**: `beatahead-ml-service`
- **Region**: `us-central1` (or region closest to Vercel production edge)
- **Container Port**: `8080` (mapped via `$PORT` environment variable)
- **CPU Allocation**: `1 vCPU`
- **Memory Allocation**: `512 MiB` (sufficient for ~60 MB runtime footprint)
- **CPU Throttling**: "CPU is only allocated during request processing" (reduces idle cost)
- **Concurrency**: `80` requests per instance
- **Min Instances**: `0` (for cost efficiency, cold start ~1.6s) or `1` (for zero cold-start latency)
- **Max Instances**: `10` (protects against denial-of-service / runaway billing)
- **Execution Environment**: Second generation (Linux container runtime)

8.2 Authentication & Security Policy

- **Ingress**: Internal + Cloud Load Balancing (or Public with IAM)
- **Authentication**: Require authentication (Service-to-service IAM).
- Vercel Next.js backend generates an OpenID Connect (OIDC) identity token using Google Service Account credentials.
- Next.js sends `Authorization: Bearer <ID_TOKEN>` on each `/predict` call.
- Cloud Run automatically verifies the cryptographic signature of the token before forwarding the request to the container.

8.3 CLI Deployment Commands (For Future Execution)

```
# 1. Build and tag image in Google Artifact Registry
gcloud builds submit --tag us-docker.pkg.dev/PROJECT_ID/beatahead-repo/beatahead-ml-service:1.0.0

# 2. Deploy to Cloud Run
gcloud run deploy beatahead-ml-service \
  --image us-docker.pkg.dev/PROJECT_ID/beatahead-repo/beatahead-ml-service:1.0.0 \
  --platform managed \
  --region us-central1 \
  --port 8080 \
  --memory 512Mi \
  --cpu 1 \
  --min-instances 1 \
  --max-instances 10 \
  --no-allow-unauthenticated
```

9. Next.js Integration Update Preparation

The Next.js integration layer (`C:\ischemic\src\app\api\ml\predict\route.ts`) is designed to seamlessly consume the Cloud Run service URL via standard environment configuration:

Configuration in `.env.production` (Vercel)

```
ML_SERVICE_URL="https://beatahead-ml-service-xyz.a.run.app"
```

Staging vs. Production Toggle

- **Local Staging**: Omits `ML_SERVICE_URL` or defaults to `http://127.0.0.1:8000`, using local daemon or CLI fallback.
- **Cloud Production**: Defines `ML_SERVICE_URL="https://..."`. Next.js routes all traffic to the containerized microservice over HTTPS with a 2,000 ms timeout.

10. Full Regression & Build Verification

The complete regression suite was executed across all testing tiers:



11. Deployment Status & Absolute Stop

> [!IMPORTANT] **PHASE 12 COMPLETED — DEPLOYMENT REMATION HALTED**
>
> In accordance with Task 12 instructions: - Container packaging, dependency manifest, and test suite are finalized. - Google Cloud Run deployment has **NOT** been triggered. - Vercel production deployment has **NOT** been triggered. - DNS has **NOT** been modified. - The frozen XGBoost model binary and threshold (`0.156742`) remain completely unchanged.